

High-Performance Markup Typography on WebAssembly

Abstract

Modern document preparation systems demand both instant feedback and high typographical precision. While existing tools like LaTeX and Beamer provide excellent output, their compilation latency and complex dependencies hinder cloud-native interactive workflows. Typst introduces a new paradigm by compiling directly to WebAssembly, enabling near-instant feedback and seamless integration with modern web applications. In this work we present **moon-typst**, an end-to-end typographical compiler implemented entirely in the MoonBit programming language. We demonstrate support for advanced features such as streaming box layout and dual-target vector generation.

Mathematical Foundation

The vertical spacing distribution model follows standard typographical baseline kinetics

$$E = mc^2$$

Furthermore, inline formulations such as $f(x) = x^2 + 1$ and fraction derivations

$$\frac{a+b}{c+d}$$

maintain harmonic font metrics and balanced baseline alignments across heterogeneous output devices.

Architecture Overview

The system pipeline consists of three decoupled phases

- 1. Lexical Tokenization & Parsing: Recursive descent parser yielding lossless AST nodes
- 2. Geometric Layout Engine: Bounding box calculation, font metric estimation, and page break partitioning
- 3. Dual-Target Vector Renderer: SVG 1.1 vector paths and semantic HTML5 DOM structures

Empirical Results

Initial micro-benchmarks indicate an order-of-magnitude throughput improvement over legacy scripting environments.